SAVITRIBAI PHULE PUNE UNIVERSITY

A REPORT ON

Heart Disease Prediction & Analysis

SUBMITTED TOWARDS THE PARTIAL FULFILLMENT OF THE REQUIREMENTS OF

BACHELOR OF ENGINEERING

(Artificial Intelligence and Data Science)

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Is a bonafide work carried out by Students under the supervision of **Mrs. Mital Kadu** and it is submitted towards the partial fulfillment of the requirement of a Bachelor of Engineering (AI&DS Engineering).

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1. Introduction

Cardiovascular diseases remain one of the leading causes of mortality worldwide, accounting for millions of deaths annually. Early detection of heart disease can significantly improve patient outcomes by enabling timely clinical intervention. However, accurately diagnosing heart disease using clinical and diagnostic parameters is a challenging task even for experienced medical professionals, owing to the complex interplay of multiple physiological factors.

Machine learning provides a powerful data-driven framework to assist clinicians in early heart disease detection. By learning patterns from historical patient records, including attributes such as age, cholesterol levels, blood pressure, chest pain type, and electrocardiographic findings, ML models can identify high-risk patients with a high degree of accuracy and consistency.

1.1 Motivation

Traditional clinical diagnosis relies heavily on physician expertise and subjective interpretation of test results. This approach is susceptible to human error, especially in resource-constrained healthcare settings. An automated predictive system built on validated clinical data can serve as a reliable decision-support tool, reducing misdiagnosis rates and facilitating preventive care at scale.

1.2 Objectives

The primary objectives of this project are:

- To develop an end-to-end machine learning pipeline for heart disease prediction using structured clinical data.
- To perform thorough data preprocessing including duplicate removal, missing value treatment, and outlier mitigation.
- To conduct comprehensive exploratory data analysis to uncover clinically meaningful patterns in the dataset.
- To evaluate and compare multiple classification algorithms including Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbours using standard performance metrics.
- To deploy the best-performing model as an interactive web application using Streamlit for real-world clinical usability.

1.3 Scope of the Project

This project covers the full machine learning lifecycle from raw data ingestion to model deployment. The scope includes data cleaning, feature analysis, model selection, and an interactive web-based frontend. The system is designed to serve as a decision support tool for clinical practitioners and health data analysts, enabling real-time risk assessment without requiring technical expertise.

2. Background and Related Work

A substantial body of research has addressed the problem of heart disease prediction using various statistical and machine learning approaches. This chapter surveys the foundational methodologies and situates the present work within the existing literature.

2.1 Traditional Statistical Methods

Early computational approaches to disease prediction relied primarily on logistic regression and linear discriminant analysis. These models provided interpretable outputs and remained computationally tractable on limited hardware. However, their reliance on linearity assumptions constrained performance when faced with the complex, non-linear feature interactions inherent in clinical data.

2.2 Machine Learning in Medical Diagnosis

The advancement of ensemble learning methods, particularly Decision Trees, Random Forests, and Gradient Boosting, significantly improved classification performance on clinical datasets. These techniques naturally handle categorical and continuous features simultaneously, tolerate limited missing data, and capture interaction effects without requiring explicit feature engineering for every relationship.

The UCI Heart Disease Dataset from the Cleveland Clinic Foundation has emerged as a widely adopted benchmark, with numerous published studies reporting classification accuracies between 78% and 90% using various ML approaches. The present project utilizes a curated variant of this benchmark dataset.

2.3 Feature Importance in Cardiac Risk

Multiple studies have identified chest pain type (cp), maximum heart rate achieved (thalach), exercise-induced angina (exang), and ST depression (oldpeak) as among the most discriminative features for heart disease classification. The correlation structure of these variables closely mirrors findings from clinical cardiology literature, providing a degree of external validity to ML-based risk scores.

2.4 Deployment of Clinical ML Models

Recent trends in clinical informatics have moved beyond model development toward deployment via user-friendly web interfaces. Streamlit has emerged as a widely adopted framework for rapid prototyping of ML-backed clinical decision support tools, enabling non-technical clinical staff to interact with predictive models without requiring programming expertise.

3. Problem Formulation

The heart disease prediction task is formally defined as a binary supervised classification problem. Given a structured feature vector of clinical measurements recorded for a patient, the objective is to predict whether that patient has heart disease (positive class, target = 1) or does not (negative class, target = 0).

3.1 Formal Definition

Let X represent an n-dimensional feature vector of clinical measurements for a given patient. The classification function f maps this feature space to a discrete binary output Y in {0, 1}, where 1 denotes presence of heart disease and 0 denotes absence. The learning objective minimizes binary cross-entropy loss over a labeled training set of N patient records.

3.2 Key Challenges

The problem presents several distinct analytical challenges:

- Class Imbalance: Although the cleaned dataset is reasonably balanced between positive and negative cases, slight imbalances can bias classifier thresholds and must be accounted for during evaluation.
- Multicollinearity: Several clinical features such as cholesterol, resting blood pressure, and age exhibit partial correlation, potentially inflating coefficient estimates in linear models.
- Outlier Sensitivity: Raw clinical values can include physiologically extreme measurements that distort model training if not appropriately treated during preprocessing.
- Feature Interpretability: For clinical deployment, it is important that model decisions remain explainable to medical professionals, favoring models with interpretable feature weights over black-box approaches.

3.3 Evaluation Metrics

Given the clinical significance of minimizing false negatives (missed diagnoses), model performance is assessed using a comprehensive set of metrics:

- Accuracy: Overall proportion of correctly classified patients.
- Precision: Proportion of predicted positive cases that are truly positive, minimizing false alarms.
- Recall (Sensitivity): Proportion of actual positive cases correctly identified, minimizing missed diagnoses.
- F1 Score: Harmonic mean of Precision and Recall, balancing both error types.
- ROC-AUC: Area Under the Receiver Operating Characteristic Curve, reflecting overall discriminative power across all classification thresholds.

4. Dataset Description

The project utilizes the Heart Disease dataset sourced from the UCI Machine Learning Repository (Cleveland Clinic Foundation). This dataset is one of the most widely cited benchmarks in cardiovascular ML research and contains structured clinical measurements from real patient records.

4.1 Raw Dataset Structure

The primary data file heart_disease_data.csv contains 1,025 rows and 14 columns, comprising 13 clinical features and one binary target variable indicating presence or absence of heart disease. After removing 723 duplicate records, the cleaned dataset retains 302 unique patient records, each representing an independent clinical observation.

4.2 Feature Descriptions

Feature	Description
age	Age of the patient in years
sex	Biological sex (1 = male, 0 = female)
cp	Chest pain type (0 = typical angina, 1 = atypical, 2 = non-anginal, 3 = asymptomatic)
trestbps	Resting blood pressure in mm Hg at hospital admission
chol	Serum cholesterol level in mg/dl
fbs	Fasting blood sugar > 120 mg/dl (1 = true, 0 = false)
restecg	Resting electrocardiographic results (0, 1, or 2)
thalach	Maximum heart rate achieved during stress test
exang	Exercise-induced angina (1 = yes, 0 = no)
oldpeak	ST depression induced by exercise relative to rest
slope	Slope of the peak exercise ST segment (0, 1, or 2)
ca	Number of major vessels colored by fluoroscopy (0-3)
thal	Thalassemia type (1 = normal, 2 = fixed defect, 3 = reversible defect)
target	Diagnosis: 1 = Heart disease present, 0 = Absent (output variable)

Table 4.1: Complete feature descriptions for the Heart Disease dataset

4.3 Target Variable Distribution

The cleaned dataset of 302 patients is reasonably well-balanced between positive cases (heart disease present) and negative cases (heart disease absent). This approximate balance reduces the need for aggressive resampling strategies and allows standard accuracy metrics to serve as meaningful evaluation criteria alongside sensitivity-specific measures.

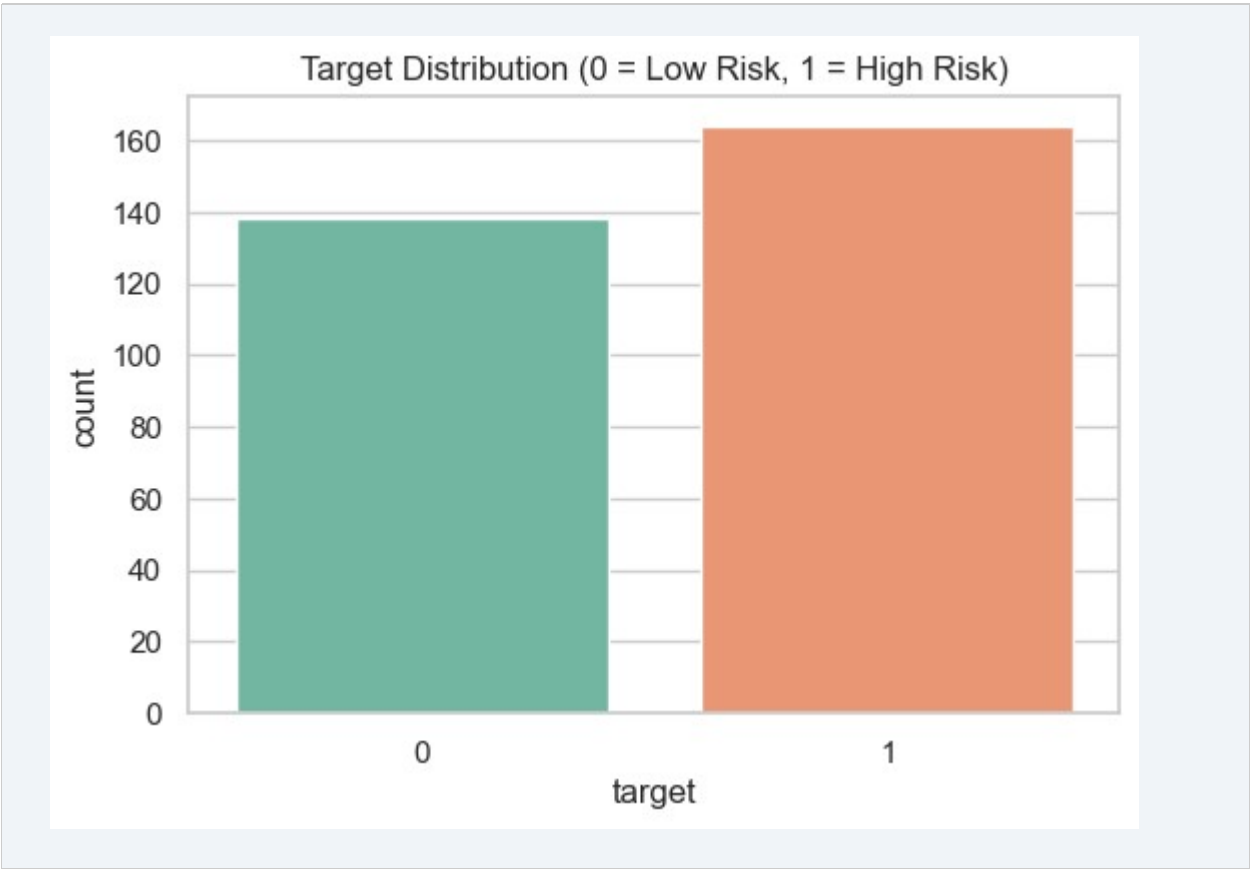

Figure 4.1: Distribution of target variable — 0 = No Heart Disease, 1 = Heart Disease Present

5. Data Preprocessing

Data preprocessing is a critical foundation for any reliable machine learning pipeline. Raw medical datasets frequently contain noise, duplicated records, and statistical outliers that can significantly degrade model performance if left unaddressed. This chapter describes the complete preprocessing workflow applied to the heart disease dataset, as implemented in 01_data_preprocessing.ipynb.

5.1 Initial Inspection

The raw dataset heart_disease_data.csv was loaded and subjected to structural inspection. It comprised 1,025 rows across 14 columns. Importantly, no missing (null) values were detected in any column, eliminating the need for imputation strategies. However, a substantial number of duplicate patient records were subsequently identified.

5.2 Duplicate Record Removal

An initial audit revealed 723 duplicate rows, representing a duplication rate of approximately 70.5% of the raw dataset. Retaining these duplicate records would introduce severe bias into model validation by allowing identical patient profiles to appear in both training and test splits, artificially inflating reported performance metrics. All duplicate rows were therefore dropped, reducing the dataset to 302 unique patient records.

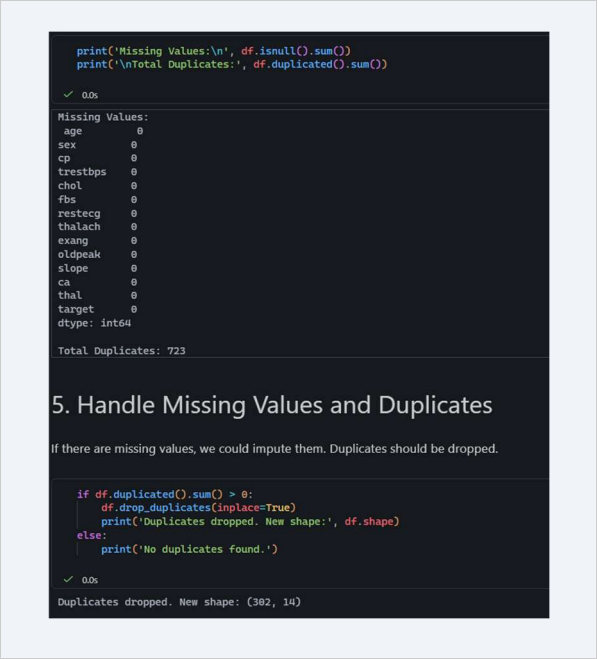

Figure 5.1: Dataset shape reduction following duplicate row removal (1,025 to 302 records)

5.3 Outlier Treatment via IQR Capping

Continuous clinical features such as serum cholesterol (chol) and resting blood pressure (trestbps) contained physiologically extreme values lying beyond plausible clinical ranges. These outliers can disproportionately distort coefficient estimates in linear models and affect distance calculations in instance-based methods like KNN.

Outliers were treated using Interquartile Range (IQR) boundary capping rather than deletion. For each numerical feature, any value exceeding Q3 + 1.5 x IQR was capped to the upper bound, and any value falling below Q1 - 1.5 x IQR was capped to the lower bound. This method preserves the full record count while containing extreme values within statistically justifiable limits.

5.4 Serialization

The fully cleaned and processed dataset comprising 302 rows and 14 columns was serialized as heart_disease_cleaned.csv. This serialization step ensures reproducibility and data consistency across the subsequent EDA and model training stages, as all downstream notebooks load exclusively from this verified clean baseline.

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis transforms the cleaned dataset into actionable clinical insights by systematically visualizing feature distributions, inter-variable relationships, and correlations with the target. This chapter summarizes the key findings from 02_EDA.ipynb, which served as the analytical foundation for feature interpretation during model training.

6.1 Univariate Analysis

Univariate histograms were generated for all continuous features. Patient ages in the dataset predominantly fall between 40 and 65 years, reflecting the clinical reality that cardiovascular risk escalates significantly with advancing age. Maximum heart rate (thalach) shows a wide distribution spanning approximately 70 to 200 beats per minute across the patient cohort.

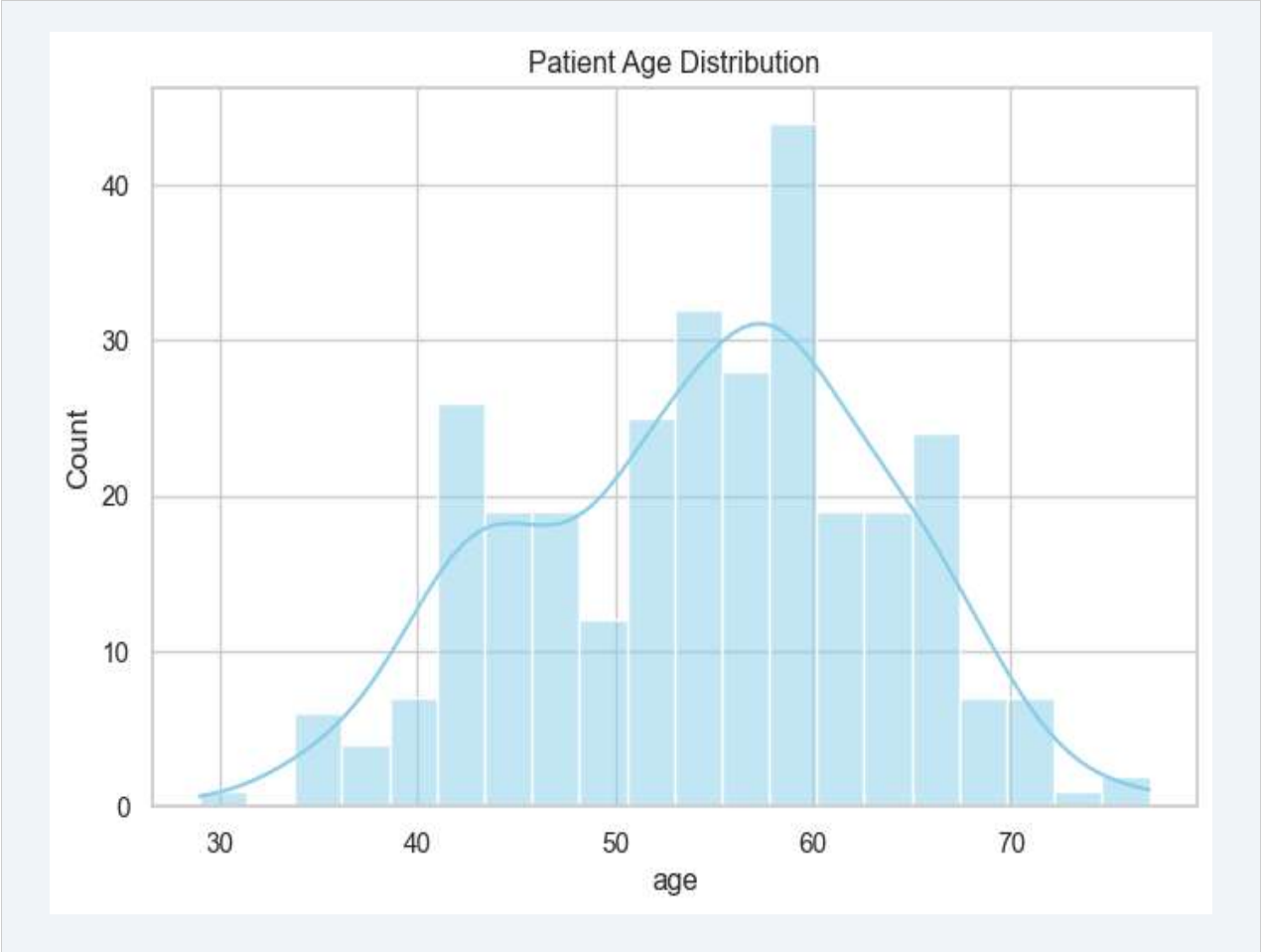

Figure 6.1: Distribution of patient ages in the cleaned dataset (predominant range: 40-65 years)

6.2 Bivariate Analysis: Features vs. Target

Bivariate analysis examined how individual features differ between patients with and without heart disease, revealing several clinically significant patterns.

6.2.1 Gender vs. Target

Male patients (1) had the highest count in both categories, especially in the low-risk group. Female patients (0) showed more cases in the high-risk group than low-risk. This suggests heart disease was more frequent overall in males, while females had a relatively higher risk proportion in this dataset.

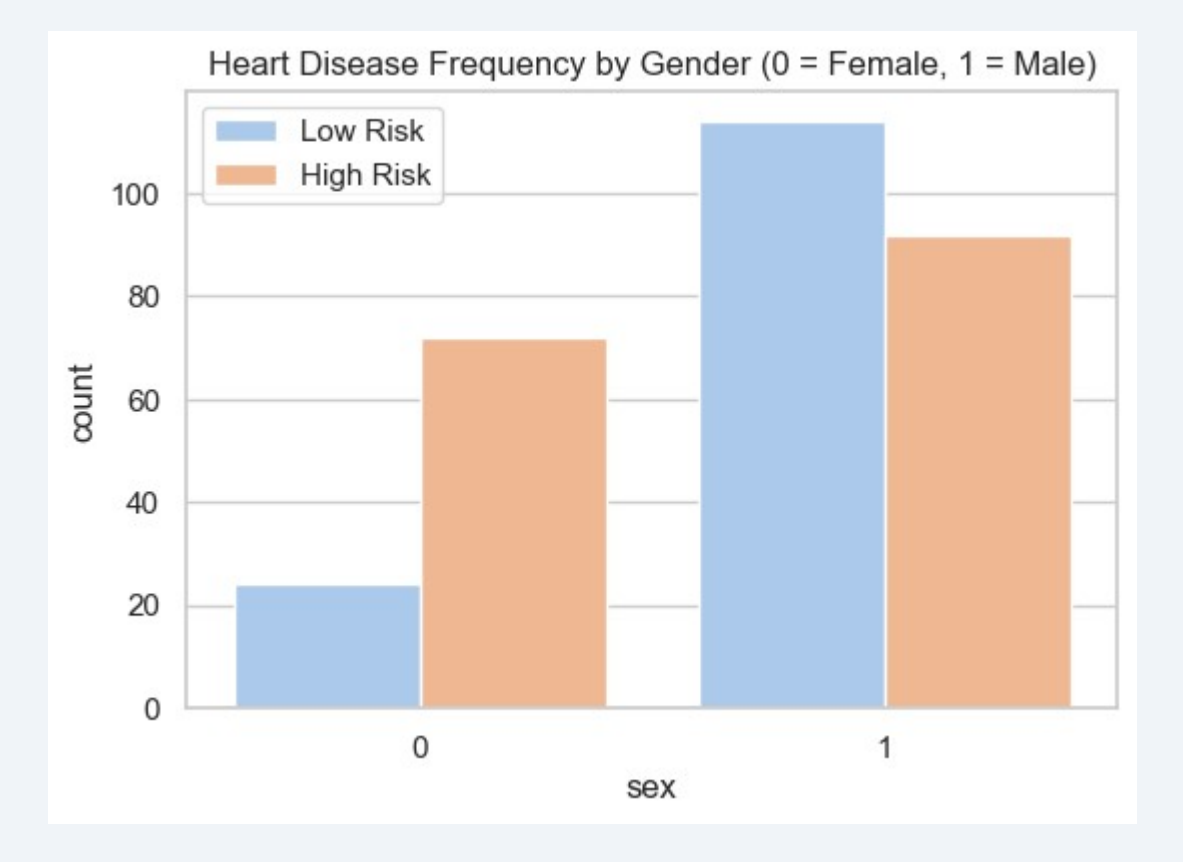

Figure 6.2: The relation between gender and Heart risks

6.2.2 Chest Pain Type (cp) vs. Target

Analysis of chest pain type revealed an important clinical finding: patients presenting with non-typical anginal pain (cp = 2) and asymptomatic presentations (cp = 3) had higher rates of positive heart disease diagnosis compared to those with typical anginal pain (cp = 0). This pattern is consistent with clinical literature suggesting atypical presentations are frequently associated with significant underlying coronary artery disease.

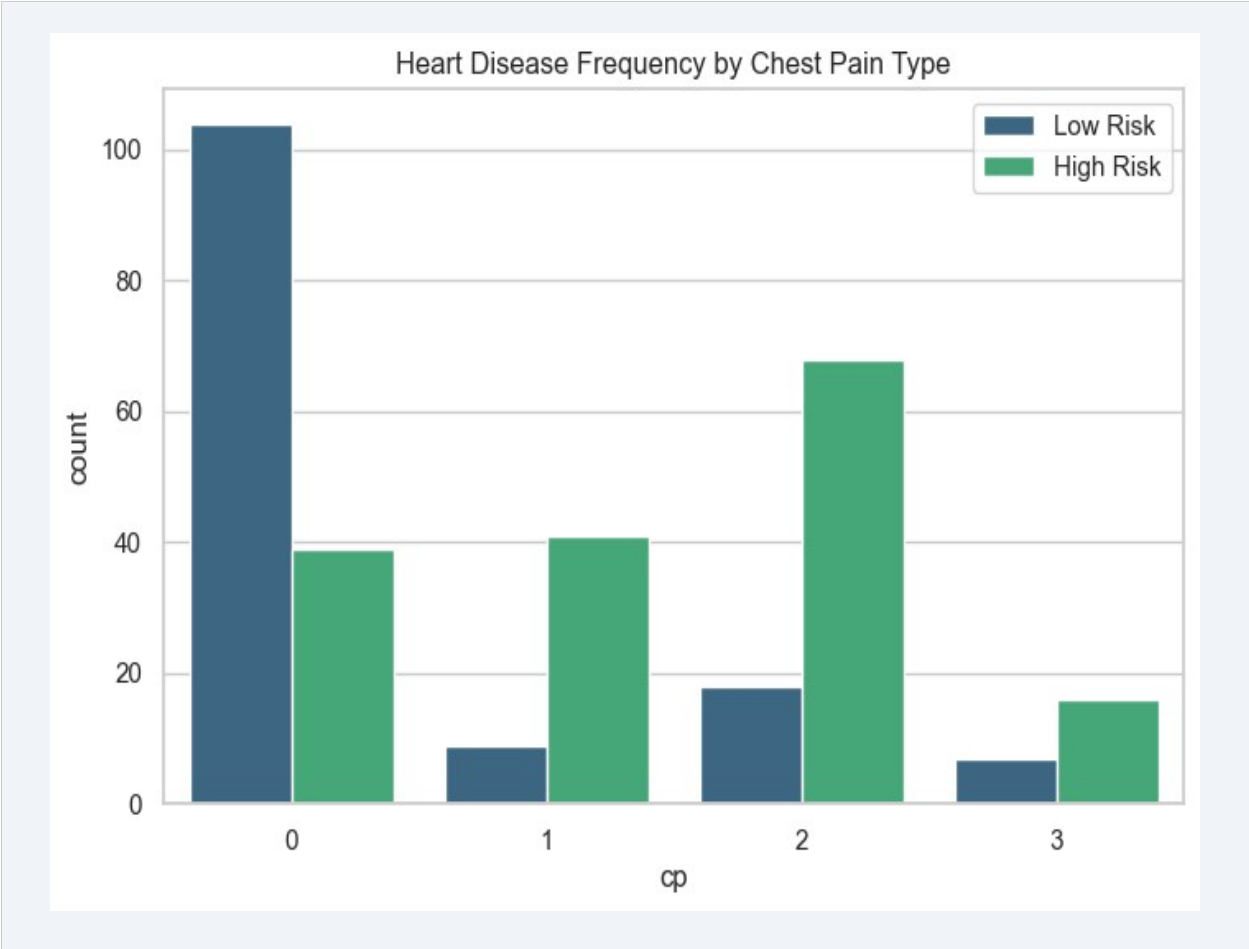

Figure 6.3: Frequency of chest pain types stratified by heart disease diagnosis

6.3 Correlation Analysis

A Pearson correlation heatmap was computed across all 13 features and the binary target variable. The following key correlations with the target were identified:

- cp (chest pain type): Strong positive correlation - non-typical pain is associated with higher disease likelihood.
- thalach (max heart rate): Positive correlation - higher heart rate associates with disease presence.
- slope (ST slope): Positive correlation - upsloping ST segments associate with disease presence.
- exang (exercise angina): Strong negative correlation - exercise-induced angina inversely associates with the positive diagnosis in this dataset.
- oldpeak (ST depression): Negative correlation - greater ST depression associates with lower disease likelihood in this cohort.

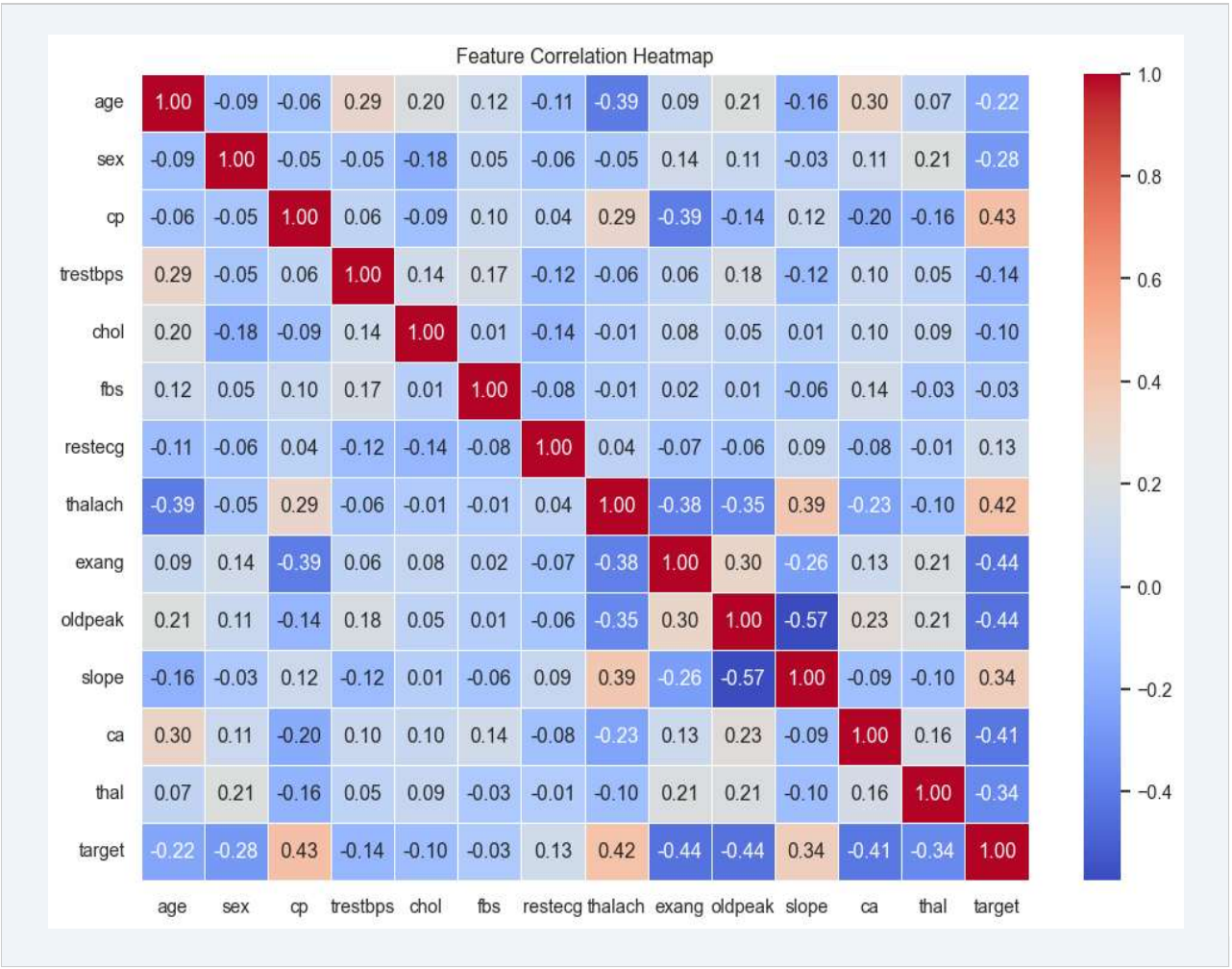

Figure 6.4: Pearson correlation heatmap across all clinical features and the target variable

7. Proposed Methodology

The project follows a structured end-to-end pipeline comprising four sequential phases: data preprocessing, exploratory analysis, model training and evaluation, and interactive deployment. Each phase is implemented as a discrete, independently executable Jupyter Notebook, enabling modular reproduction and validation of individual stages without rerunning the full pipeline.

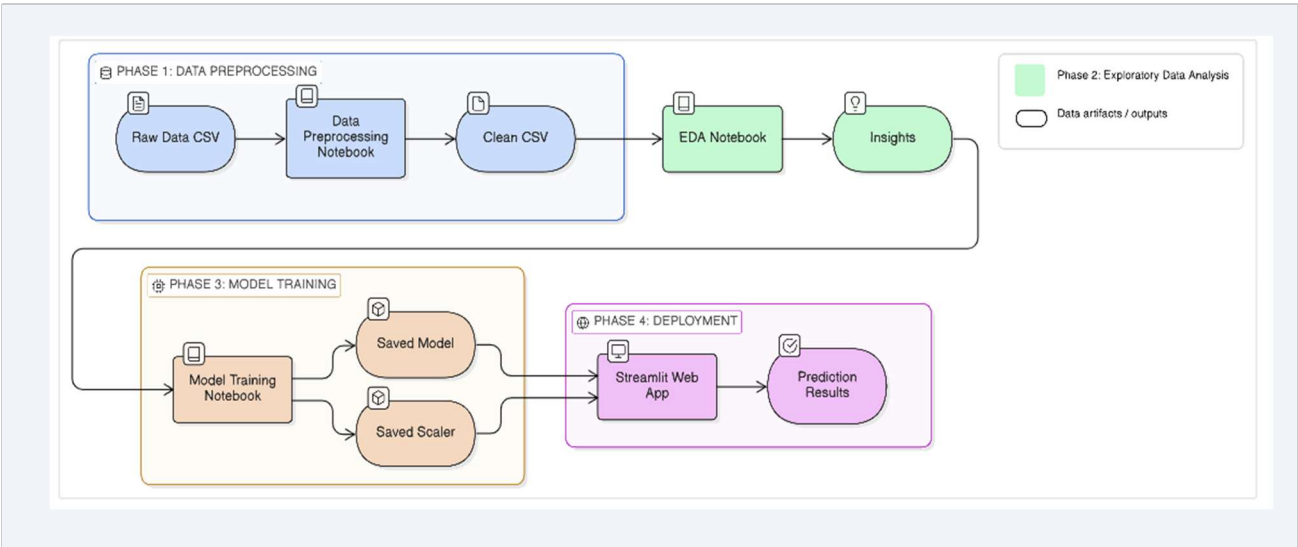

Figure 7.1: End-to-end project pipeline architecture from raw data to deployed application

7.1 Phase 1: Data Preprocessing (01_data_preprocessing.ipynb)

The raw CSV is loaded, structurally validated, duplicate records are removed, and continuous features are subjected to IQR-based outlier capping. The cleaned dataset of 302 rows is serialized as heart_disease_cleaned.csv for downstream use. Full preprocessing details are provided in Chapter 5.

7.2 Phase 2: Exploratory Data Analysis (02_EDA.ipynb)

The cleaned dataset is loaded and subjected to comprehensive univariate, bivariate, and correlation analysis. Visualizations include histograms, box plots, count plots, and a full-feature correlation heatmap. Insights from this phase directly informed feature interpretation and provided qualitative validation of the model's learned coefficients. Full EDA details are covered in Chapter 6.

7.3 Phase 3: Model Training and Evaluation (03_model_training_evaluation_prediction.ipynb)

The cleaned dataset is split using a stratified 80/20 train-test ratio to preserve class balance across both partitions. Feature standardization is applied using StandardScaler and the fitted scaler state is serialized as scaler.pkl for consistent inference at deployment time. Four classification algorithms are trained and benchmarked simultaneously before the best-performing model is exported as best_heart_disease_model.pkl. Full training details are covered in Chapter 8.

7.4 Phase 4: Streamlit Deployment (app.py)

The serialized model and scaler are loaded into a Streamlit web application providing a dual-tab interactive interface. Tab 1 provides real-time clinical risk prediction for individual patients. Tab 2 provides an interactive EDA dashboard with live Plotly visualizations. Full application details are provided in Chapter 9.

8. Model Training and Evaluation

This chapter presents the complete model selection workflow, detailing the train-test split strategy, feature scaling methodology, individual model performance results, final model selection rationale, and end-to-end inference validation.

8.1 Train/Test Split

The 302-record cleaned dataset was split using stratified sampling with a fixed random seed to ensure reproducibility. Stratification ensures that the approximate class distribution of the target variable is preserved in both training and test partitions, preventing accidental class bias in evaluation.

- Training Set: 241 samples (80%)
- Test Set: 61 samples (20%)

8.2 Feature Scaling

Clinical features occupy vastly different numerical scales. For example, serum cholesterol ranges into the hundreds (mg/dl) while ST depression (oldpeak) takes decimal values near zero. Distance-based models such as KNN and gradient-based models such as Logistic Regression are highly sensitive to these scale disparities. StandardScaler was applied to the training features, transforming each feature to zero mean and unit variance. The fitted scaler was then applied identically to the test set without refitting, to prevent data leakage. The scaler state was serialized to scaler.pkl for use at inference time in the deployed Streamlit application.

8.3 Model Evaluation Results

Four classification algorithms were trained on the standardized training set and evaluated on the held-out test set. Results are summarized below:

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	80.33%	80.00%	84.85%	82.35%	87.23%
Decision Tree	80.33%	81.82%	81.82%	81.82%	80.19%
KNN	80.33%	80.00%	84.85%	82.35%	84.47%
Random Forest	75.41%	76.47%	78.79%	77.61%	86.47%

Table 8.1: Comparative performance metrics of all evaluated classification models on the held-out test set

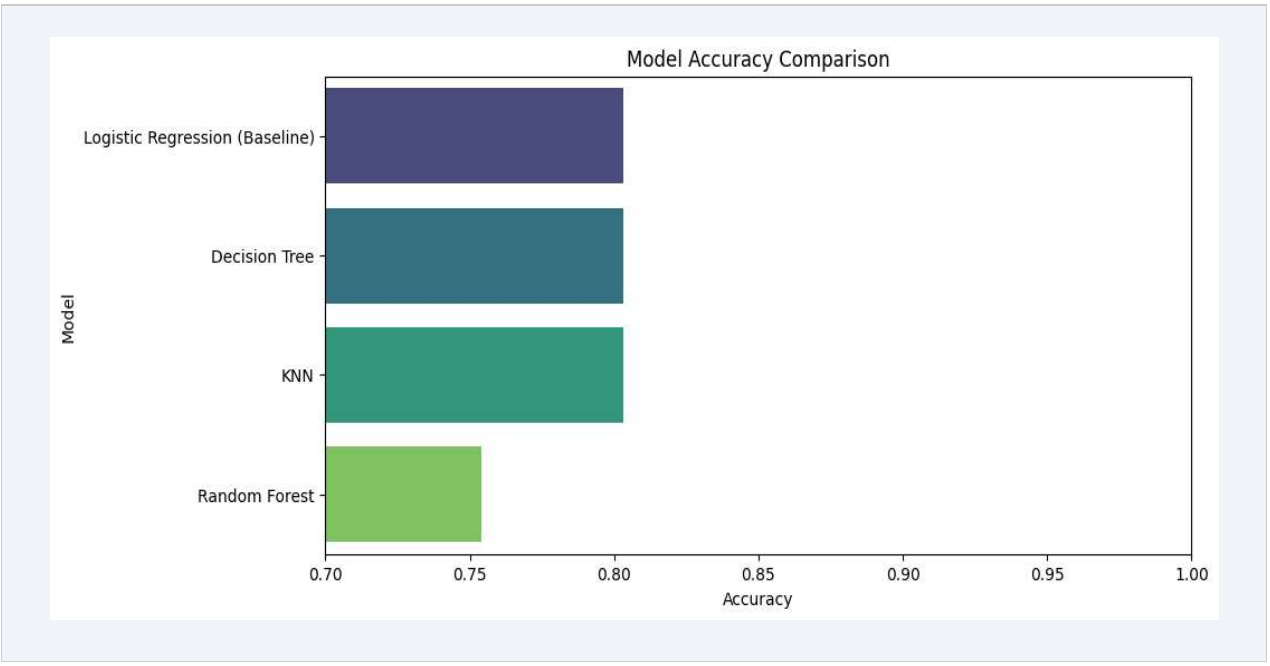

Figure 8.1: Side-by-side comparison of classifier performance metrics across all models

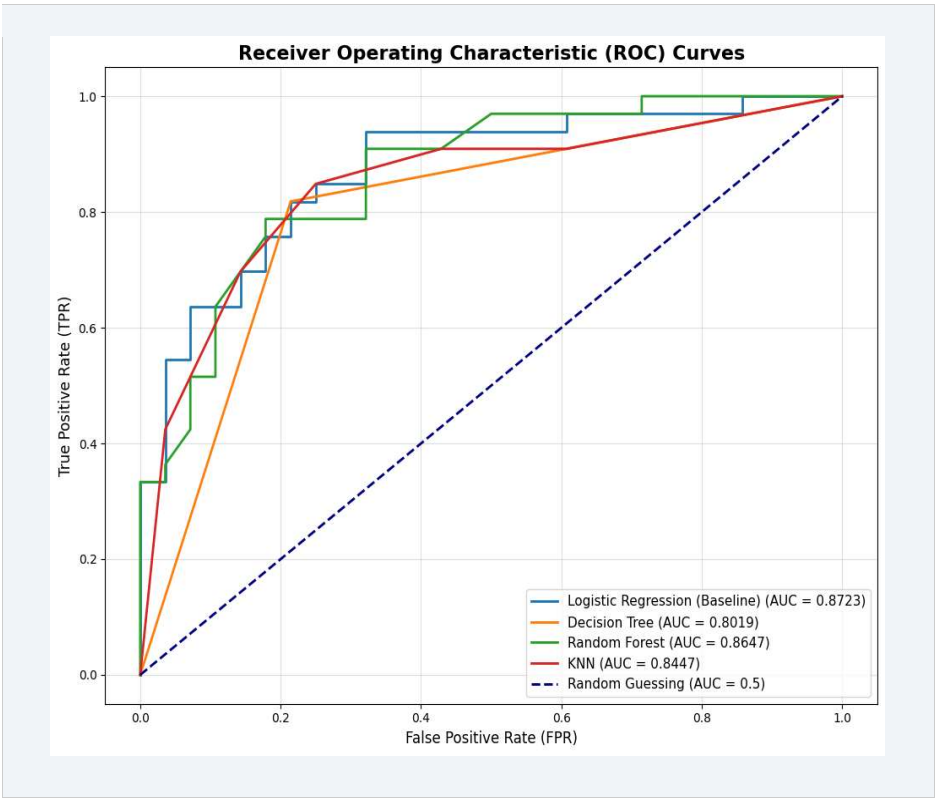

Figure 8.2: ROC curves for all classifiers higher curve area indicates superior discriminative ability

8.4 Final Model Selection: Logistic Regression

All three top models, Logistic Regression, Decision Tree, and KNN, achieved identical accuracy of 80.33% on the test set. Logistic Regression was selected as the final production model based on two decisive criteria:

- Superior ROC-AUC (87.23%): The ROC-AUC score reflects model performance across all classification thresholds, making it a more robust indicator of generalization than accuracy alone. Logistic Regression's AUC exceeded KNN by 2.76 percentage points and Decision Tree by 7.04 percentage points.
- Clinical Interpretability: Logistic Regression produces calibrated probability estimates and transparent feature coefficients, enabling clinicians to understand the contribution of individual variables to predicted risk. This interpretability is a critical requirement for clinical trust and regulatory compliance in medical AI applications.

8.5 Inference Pipeline Validation

To validate end-to-end inference readiness prior to deployment, a mock patient profile was constructed programmatically and passed through the complete inference pipeline: feature standardization via the serialized scaler.pkl followed by probabilistic prediction via best_heart_disease_model.pkl. The model returned a 77.59% probability of High Risk, confirming that the serialized artifacts function correctly and consistently for real-time prediction within the Streamlit application.

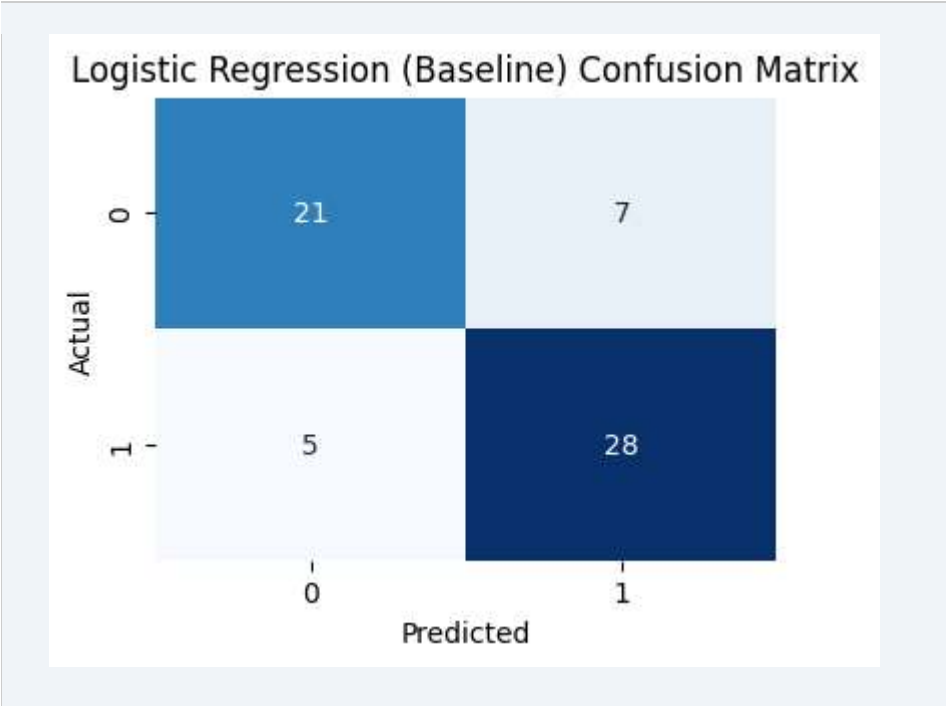

Figure 8.3: Logistic Regression confusion matrix on held-out test set

9. Streamlit Web Application

The final deliverable of this project is an interactive, browser-based web application built with Streamlit that enables clinical practitioners and data analysts to interact with the trained model without requiring any programming expertise. The application exposes the full predictive pipeline through a clean dual-tab interface accessible in any modern web browser.

9.1 Application Architecture

The app.py script loads the serialized scaler (scaler.pkl) and model (best_heart_disease_model.pkl) on startup into memory. User-submitted clinical inputs are captured via Streamlit UI widgets, scaled using the loaded scaler to match the distribution seen during training, and passed to the model for inference. Results are rendered immediately in the browser without any page reload or external API call.

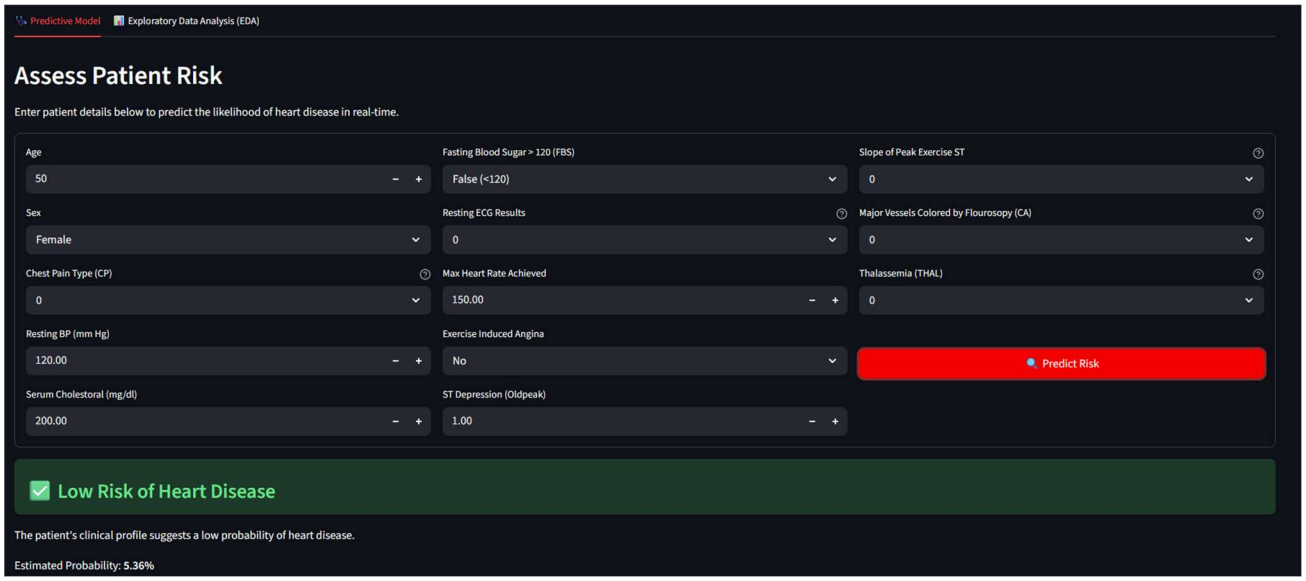

Figure 9.1: Streamlit application Tab 1 — Real-time Heart Disease Risk Prediction Interface

9.2 Tab 1: Predictive Model Interface

The first tab provides a structured input form covering all 13 clinical features required by the model:

- Demographic inputs: Age , Sex .
- Clinical measurements: Resting Blood Pressure (trestbps), Serum Cholesterol (chol), Fasting Blood Sugar threshold flag (fbs).
- Diagnostic parameters: Resting ECG result (restecg), Maximum Heart Rate (thalach), Exercise-Induced Angina (exang), ST Depression (oldpeak), ST Slope (slope).
- Imaging and lab results: Number of major vessels colored by fluoroscopy (ca), Thalassemia type (thal), Chest Pain Type (cp).

Upon clicking the Predict button, the backend passes all inputs through scaler.pkl for standardization and invokes best_heart_disease_model.pkl to produce a binary classification alongside a calibrated probability estimate. The result is displayed as a risk level (High Risk or Low Risk) with the associated confidence percentage rendered prominently on-screen.

9.3 Tab 2: Interactive EDA Dashboard

The second tab embeds live Plotly interactive charts directly within the browser, allowing users to explore the underlying dataset without requiring a local Python environment or Jupyter installation. Users can hover over data points for exact values, zoom into regions of interest, and toggle series visibility. Available visualizations include correlation heatmaps, age distribution histograms, chest pain type frequency breakdowns stratified by diagnosis status, and feature-wise box plots for continuous variables grouped by target class.

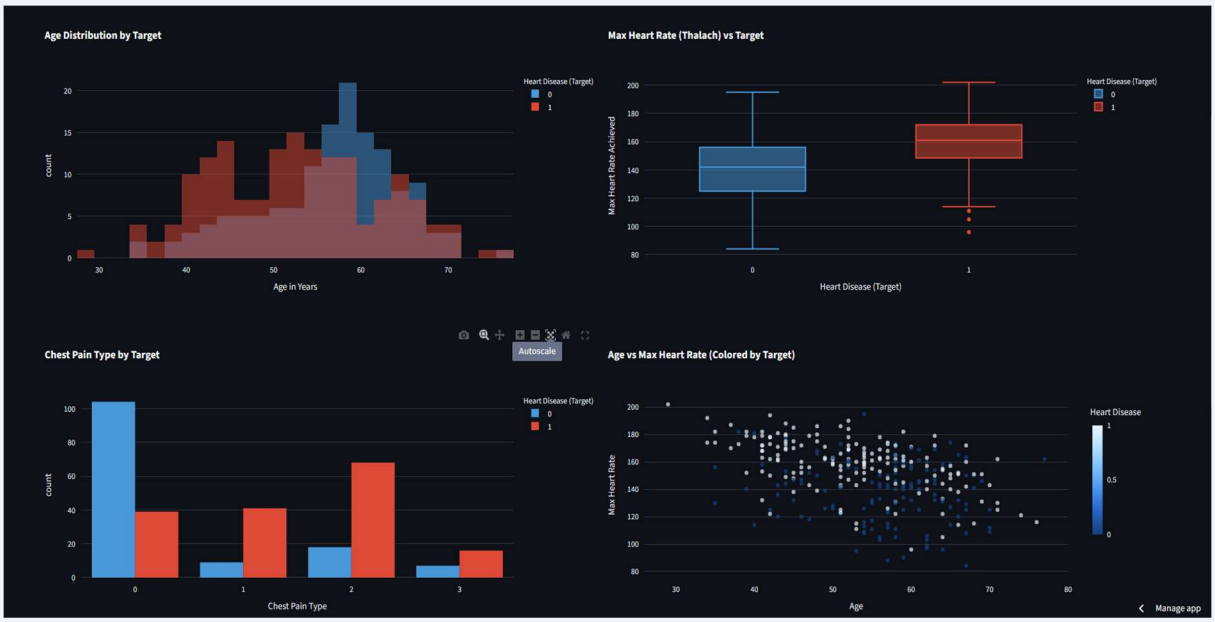

Figure 9.2: Streamlit application Tab 2 — Interactive EDA Dashboard with Plotly visualizations

10. Results and Discussion

This chapter synthesizes the key quantitative outcomes of the project and interprets their clinical and technical significance. The discussion contextualizes model performance within the broader landscape of heart disease prediction research and identifies directions for future improvement.

10.1 Summary of Results

The complete ML pipeline achieved the following primary outcomes:

- Dataset Quality: Removal of 723 duplicate records (70.5% of raw data) produced a reliable 302-patient clean dataset, significantly improving the integrity of model validation and reducing overfitting risk.
- Best Model Performance: Logistic Regression achieved the highest ROC-AUC of 87.23% with 80.33% accuracy and 84.85% recall on the held-out test set.
- Clinically Relevant Recall: The 84.85% recall rate means the model correctly identifies approximately 85 out of every 100 true positive heart disease patients, a clinically meaningful sensitivity level for a screening tool where missed diagnoses carry severe consequences.
- Deployment Validated: End-to-end inference through the serialized scaler and model artifacts was confirmed via a mock patient profile, producing a 77.59% high-risk probability estimate consistent with expected model behavior.

10.2 Clinical Interpretation of Key Findings

The EDA phase revealed that chest pain type (cp), maximum heart rate (thalach), and exercise-induced ST slope are the most discriminative features for heart disease classification in this dataset. These findings are consistent with established cardiovascular risk stratification frameworks, providing a measure of external validity to the learned model and lending credibility to its clinical applicability.

The logistic model's calibrated probability output is particularly valuable in a clinical context. Rather than relying solely on the model's default binary decision boundary, clinicians can apply institution-specific risk thresholds to adjust the sensitivity-specificity trade-off based on local patient population characteristics and resource constraints.

10.3 Limitations

Several limitations should be transparently acknowledged:

- Dataset Size: The cleaned dataset of 302 patients, while sufficient for exploratory modeling and academic benchmarking, is relatively small for clinical deployment. Larger multicenter datasets would improve generalizability across diverse patient demographics.
- Feature Completeness: Clinically important variables such as family history, smoking status, body mass index, and physical activity levels are absent from the dataset, limiting the comprehensiveness of the resulting risk model.
- Static Dataset: The dataset does not include temporal metadata. The extent to which the learned distribution reflects current clinical populations is therefore uncertain.
- Single Dataset Source: Validation was performed exclusively on a held-out split of the same dataset. External validation on an independent clinical cohort would be required before considering any real-world clinical deployment.

11. Conclusion

This project successfully developed and deployed an end-to-end machine learning pipeline for heart disease prediction using structured clinical data. Starting from a raw 1,025-record dataset, the pipeline progressed through rigorous data cleaning, comprehensive exploratory analysis, comparative model evaluation, and interactive web deployment, demonstrating the practical feasibility of ML as a clinical decision support tool.

Logistic Regression was selected as the production model based on its superior ROC-AUC of 87.23% and clinically favorable recall of 84.85%, alongside its inherent interpretability. The model was deployed as a dual-tab Streamlit application enabling real-time clinical risk scoring and interactive data exploration for practitioners without programming expertise.

The project demonstrates that a carefully constructed ML pipeline, even on a modestly sized clinical dataset, can yield meaningful predictive performance when grounded in rigorous preprocessing, thoughtful feature analysis, and principled model selection. The Streamlit deployment bridges the gap between model development and clinical utility by making the predictive system immediately accessible and usable.

11.1 Future Scope

- Expand the training dataset by integrating publicly available multi-institutional cardiac datasets to improve model generalizability across diverse patient demographics.
- Incorporate additional clinical risk factors such as smoking history, family history of heart disease, body mass index, and physical activity levels to enrich the predictive feature set.
- Explore advanced classification architectures including XGBoost, LightGBM, and deep neural network approaches to assess potential accuracy improvements.
- Implement SHAP explainability visualizations within the Streamlit application to provide per-prediction feature attribution, improving clinician trust and interpretability of model outputs.
- Conduct external validation of the trained model on an independent clinical dataset to assess real-world generalizability before considering any clinical deployment.
- Develop a mobile-responsive version of the application for use in primary healthcare and telemedicine settings with limited infrastructure.

12. References

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